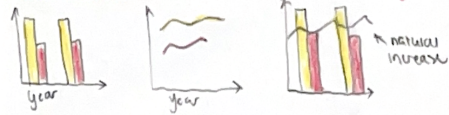


1. IDEAS

Deaths & Births

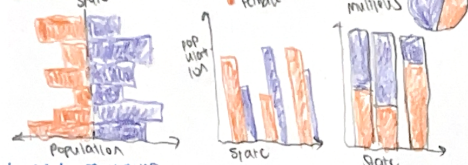


→ Can do by whole of Australia or State by State

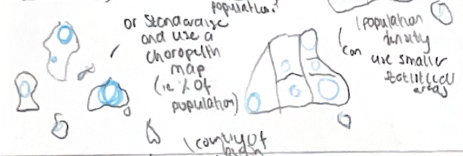
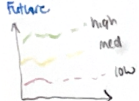
Population Overall



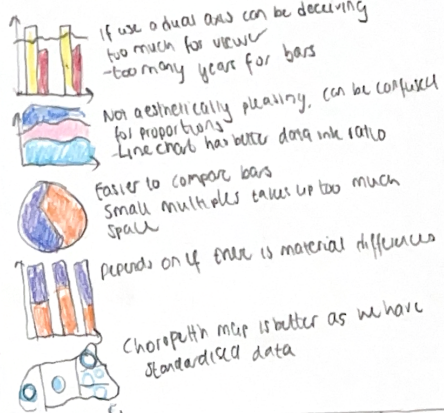
2021 Census State or age group



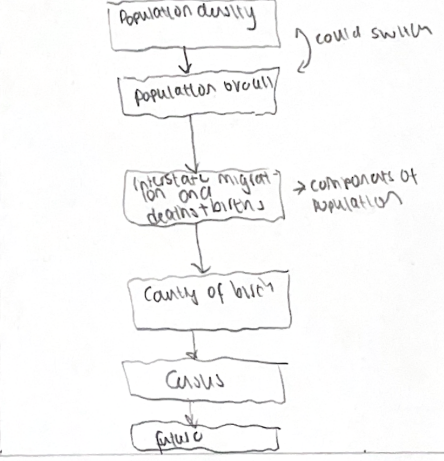
Intrastate migration



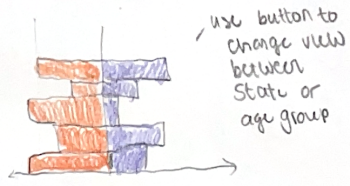
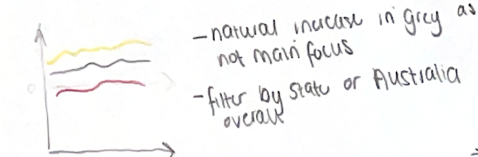
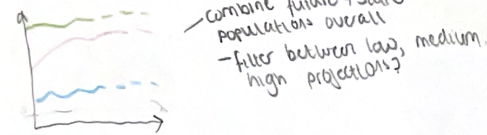
2. FILTER



3. CATEGORISE



4. COMBINE AND REFINE



5. QUESTION

- Is it possible to filter in vega lite
- Is there too much going on which reduces storytelling
- Is it better to show absolute values or relative for state?
- Better to show all of Australia & state by state?

AUSTRALIA'S POPULATION STATE BY STATE



Text

Filter

Birth & deaths

Migration interstate

Male vs female

future

Text

Title: Australia's population state by state

Author: Tanisha Wu

Date: 10/9/2024

Sheet: 2

Task: FIT3179 visualisation 2

OPERATIONS



filter by state to
select view.
Change graphs
in the bottom
half



time to change
male vs female
proportion

could also use graph to
filter by state

Annotate interesting findings
If possible use of tooltips to
exact details

FOCUS

Focus is the choropleth graph



Text

- Select a state to filter other charts on the visualisation
- Each state coloured by population density (persons/km²)
- Annotations to also have total population and the area of each state in 2023
- Need to make sure that there is enough distinction between colour brightness

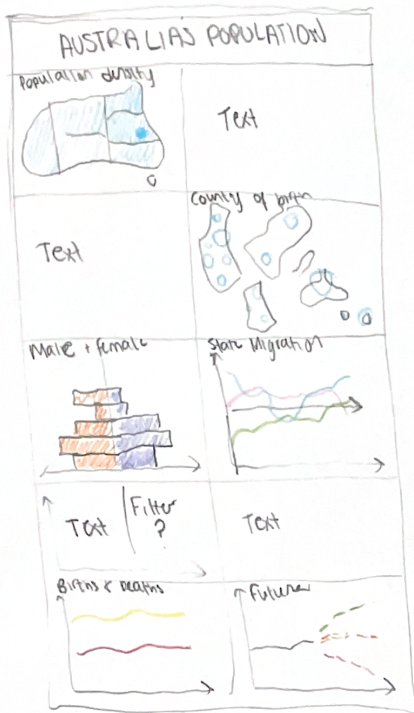
DISCUSSION

Advantages

- Not too overwhelming on viewer as only focus by state
- Clear focus
- Chore variation

Disadvantages

- Misses out on any potential insights from all of Australia
- Okay story telling
- May need separate vectorlike files for each state for each graph



OPERATIONS

Text panels to add content

→ Also could add annotations to visualisations as most are static - emphasise notable points

Potential to add a filter for male and female proportions

Also opportunity to add a filter or brush for time on state migration and births and deaths

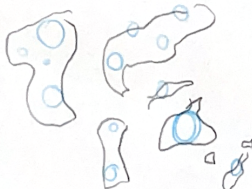
FOCUS

Main focus is two maps
Australia population density



- Choropleth map
- Population density (persons/km²)
 - Tooltip show exact numbers
 - Text for both graphs to add context

Country of birth of Australians



- Proportionate symbol map
- Australians born in that country
 - Transparent symbols for visibility
 - Tooltips over symbol to show exact amounts
 - Could have filter to major countries, major or minor group

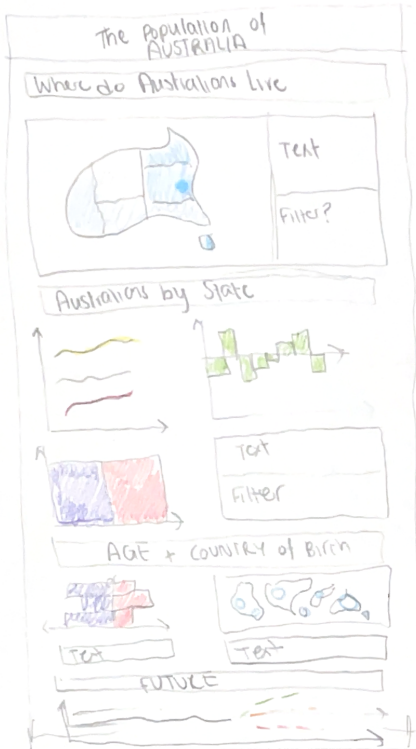
DISCUSSION

Advantages

- Clear sight lines
- Balanced layout
- Good variety of graphs
- Text adds context

Disadvantages

- Limited interaction
- Storytelling is unclear
- May not meet task requirements: MAJOR ISSUE
- Potentially too many colours



Title: The Population of Australia

Author: Tonisha Wu

Date: 10/9/2024

Sheet: 4

Task: FIT3179 visualisation 2

OPERATIONS

- Button / filter for top map to change view between choropleth map with states and smaller regional areas or change to proportional symbol map
- filter by state and maybe time for state section
- Tool tips to show details
- Click to highlight different future scenarios

FOCUS

Choropleth map showing population density (persons / km²)



- Text to outline and describe key findings
- filter to change view by state to smaller regional areas → more detail
- Potentially add a button to change to a proportional symbol map showing total population in each location

DISCUSSION

Advantages:

- Swatches help guide the reader - better storytelling
- fairly balanced with good sight lines

Disadvantages

- Interactivity still questionable
- Order of sections may not form a coherent story

The Population of Australia

Where are Australians born?



Text

change view

Where do Australians live?



Text

change view

Australians by State



Births or deaths

Text

Age and Sex of Australians



Text

The Future Population



Text

Title: The Population of Australia
Author: Tanisha Wu

Date: 11/9/2024

Sheet: 5

Task: FIT3179 visualisation 2

OPERATIONS

Buttons

- for both maps buttons can be used to change views
- button to switch between births or deaths by state

Tool-tips

- most graphs will have tooltips to show finer details
- interactive migration use tooltips to show net migration at a particular time

Legend

- for state clicking a colour will highlight that state (hopefully in both graphs)

Brushing

- highlight points on maps

FOCUS

Two main focuses: two maps



- Proportional symbol map to choropleth map
- Proportional symbol shows raw data by (country) major or minor groups
- Choropleth map shows overall proportion of Australians from (country) major or minor group
- Button to switch graphs
- Text to provide context
- Tooltips showing information



- Choropleth map to ordered choropleth map
- Both maps show population density but one shows by state, the other with more detail within states
- Button to switch between choropleth maps
- Accompanying text and tooltips for details

DETAIL

Dependencies

- R for any data wrangling
- Vega.js to create charts
- HTML to create vis using framework to embed charts

Estimated time and effort

- At time of creating sheet ~1 month until due date
- 0.1 day to manipulate data to required form
- 0.1 day for research of data on geographical locations
- 0.1 week to create charts
- 0.3 days to create vis

Specific requirements

- Longitude and latitude data for maps - also GeoJSON
- Understanding of how to add interaction with Vega.js
- formatting with CSS
- long format for vertical scrolling